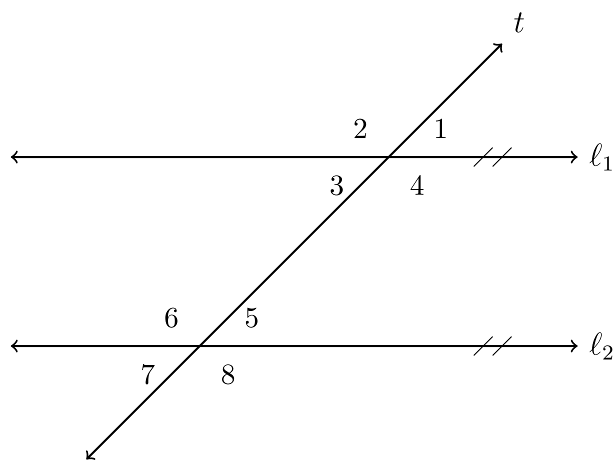
Angle Pairs Formed by a Transversal

Directions:

- Every problem on this sheet refers to the one diagram below. It shows only the numbering of the eight angles — *no measures are marked on it.* Each problem supplies its own measures.
- Name the angle pair you are using before you compute. “Corresponding” or “alternate interior” is part of the answer, not decoration.
- Give angle measures in degrees.



The one diagram for every problem. Lines ℓ_1 and ℓ_2 are parallel (matching arrowheads); line t is the transversal. No values are shown — use the measures given in each problem.

1. Name the relationship between each pair of angles. Choose from: corresponding, alternate interior, alternate exterior, same-side (co-interior), vertical, linear pair.

- $\angle 1$ and $\angle 5$ ____________________
- $\angle 4$ and $\angle 6$ ____________________
- $\angle 2$ and $\angle 8$ ____________________

2. Suppose $m\angle 1 = 112^\circ$. Find $m\angle 5$ and then $m\angle 6$. Name the pair you used for each step.

Answer: ________________

3. Suppose $m\angle 4 = 65^\circ$. Find $m\angle 6$, then find $m\angle 5$. One of these two angles is congruent to $\angle 4$ and the other is supplementary to it — the names tell you which is which.

Answer: ________________

4. $\angle 2$ and $\angle 6$ are corresponding angles, with $m\angle 2 = (3x + 10)^\circ$ and $m\angle 6 = (5x - 24)^\circ$. Write the equation the angle pair justifies, solve for x, and then give $m\angle 2$.

Answer: ________________

5. Suppose $m\angle 7 = 74°$. Find $m\angle 1$, then find $m\angle 4$. State the pair name used at each step.

Answer: ________________

6. $\angle 3$ and $\angle 6$ are same-side interior angles, with $m\angle 3 = (2x + 15)°$ and $m\angle 6 = (3x + 40)°$. Same-side interior angles are supplementary, not congruent — write the correct equation, solve for x, and give $m\angle 3$.

Answer: ________________

7. $\angle 2$ and $\angle 8$ are alternate exterior angles, with $m\angle 2 = (4x - 9)°$ and $m\angle 8 = (3x + 21)°$. Find x and $m\angle 2$. Then check your answer a second way: find $m\angle 7$ and confirm that $\angle 7$ and $\angle 8$ form a linear pair.

Answer: ________________

8. Explain why $\angle 3$ and $\angle 6$ must be supplementary whenever $\ell_1 \parallel \ell_2$. Use at least one angle-pair name and one other angle in the diagram as a stepping stone. Two or three sentences.